AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target: metZ (UniProt **Q88LD4**), *Pseudomonas putida* KT2440 (NCBITaxon:160488) **Seed hypothesis under test:** metZ has **carbon-sulfur lyase activity (GO:0016846)** — propagated by TreeGrafter/PANTHER (IEA, GO_REF:0000118) **Hypothesis slug:** function-hypothesis-go-0016846

Executive Judgment

Verdict: REFUTED — Within-superfamily mis-placement (Failure Mode 3).

The TreeGrafter/PANTHER graft that assigned **carbon-sulfur lyase activity (GO:0016846)** to metZ is **wrong**. metZ is not a lyase at all; it is a **PLP-dependent O-succinylhomoserine sulfhydrylase**, a **transferase** (EC 2.5.1.-) that catalyzes direct sulfhydrylation in methionine biosynthesis. The graft placed the query onto a structurally related but functionally *distinct* neighboring subfamily of the shared **Cys/Met-metabolism PLP-dependent (transsulfuration) fold superfamily** — specifically the cystathionine γ-lyase branch (PANTHER PTHR11808:SF80) — whose enzymes cleave carbon–sulfur bonds in the *reverse* chemical direction.

The single most decisive piece of evidence: metZ is **82.1% identical** to the experimentally characterized *P. aeruginosa* MetZ O-succinylhomoserine sulfhydrylase (P55218, EC 2.5.1.-, a transferase), but only **~36.5% identical** to human cystathionine γ-lyase (P32929, EC 4.4.1.1, the C–S lyase whose label drove the graft). The nearest *characterized* neighbor is unambiguously a transferase, and its reaction chemistry — net **addition** of sulfide to O-succinylhomoserine — is the opposite of the "elimination of hydrogen sulfide" that defines GO:0016846.

This is **not** a pseudo-enzyme (Failure Mode 2): the catalytic PLP Schiff-base lysine (K219, within the canonical **SATKF** motif) is intact and correctly spaced, confirming the protein is a fully active enzyme. And it is **not** a mere granularity problem (Failure Mode 1): the issue is not that GO:0016846 is "too broad" but that it names the **wrong reaction class** (lyase vs. transferase) in a sibling subfamily. The correct molecular-function branch is **transferase**, not lyase.

Independent Family/Function Assignment

From primary evidence, metZ (Q88LD4) is an **O-succinylhomoserine sulfhydrylase**:

> O-succinyl-L-homoserine + H_2S → L-homocysteine + succinate

This is the first (and in *P. putida*, the committed) step of L-homocysteine synthesis from O-succinylhomoserine, feeding directly into **L-methionine biosynthesis** via **direct sulfhydrylation**. The enzyme is PLP-dependent (cofactor pyridoxal 5′-phosphate, covalently bound as N6-(pyridoxal phosphate)-lysine at **K219**).

Attribute	Independent assignment
Specific MF	O-succinylhomoserine sulfhydrylase
EC class	EC 2.5.1.- (**transferase**, transferring alkyl/aryl groups other than methyl)
Recommended GO term	**GO:0016765** — transferase activity, transferring alkyl or aryl (other than methyl) groups
Characterized homolog basis	*P. aeruginosa* MetZ (P55218), 82.1% identical, experimentally characterized sulfhydrylase
Granularity vs. seed (GO:0016846)	**Different branch** — the seed term is in the lyase (EC 4) ontology branch; the correct term is in the transferase (EC 2) branch. Not "same," not "more specific," not a close sibling within one activity — a different reaction class.

Notably, QuickGO already shows Q88LD4 carrying **GO:0016765** (UniRule UR000282477) — the branch-correct transferase term — *alongside* the disputed GO:0016846. The curation task is therefore to correct a conflicting, mis-grafted lyase annotation that coexists with the correct transferase annotation.

Key Findings

Finding 1 — metZ is a PLP-dependent sulfhydrylase (transferase), not a carbon-sulfur lyase

The reviewed UniProt entry for Q88LD4, annotated via HAMAP rule **MF_02056** ("MetZ subfamily"), gives the recommended name **O-succinylhomoserine sulfhydrylase** with EC class **2.5.1.-** — unambiguously the **transferase** class. The catalytic reaction it annotates is *O-succinyl-L-homoserine* + H_2S → *L-homocysteine* + *succinate*, with PLP cofactor bound at **K219** and a pathway assignment of *L-methionine biosynthesis; L-homocysteine from O-succinyl-L-homoserine, step 1 of 1*.

Crucially, the Gene Ontology definition of the seed term **GO:0016846 (carbon-sulfur lyase activity)** describes the **elimination of hydrogen sulfide** — a β- or γ-**elimination** that *breaks* a C–S bond and *releases* sulfide. metZ does the chemical opposite: it nets the **addition** of sulfide (H_2S) across O-succinylhomoserine to build homocysteine. A reaction that consumes H_2S to form a C–S bond cannot be described by a term defined as sulfide elimination. This directional mismatch is dispositive at the level of reaction chemistry, independent of any sequence analysis.

This finding is directly supported by the primary literature that first identified and sequenced metZ:

> *"The second step is achieved by direct sulfhydrylation, involving the enzyme encoded by a metZ gene that we have identified and sequenced"* — PMID: 7704274

Direct sulfhydrylation is a γ-replacement/transferase reaction, not a lyase cleavage. This establishes the enzyme's mechanism from experimental work, not from database inference.

Finding 2 — The nearest *characterized* homolog is a transferase (82% identity); C–S lyases are distant (≤37%)

To localize the graft error, pairwise global (Needleman–Wunsch) alignments were computed between the query Q88LD4 and a panel of characterized members of the transsulfuration/Cys-Met PLP superfamily spanning both the transferase and lyase branches:

Characterized protein	Accession	Activity	EC	Class	% identity to Q88LD4
P. aeruginosa MetZ	P55218	O-succinylhomoserine sulfhydrylase	2.5.1.-	**Transferase**	**82.1%**
E. coli MetB	P00935	cystathionine γ-synthase	2.5.1.48	**Transferase**	38.0%
Human CTH	P32929	cystathionine γ-lyase	4.4.1.1	**C–S lyase**	36.5%
E. coli MetC	P06721	cystathionine β-lyase	4.4.1.8	**C–S lyase**	25.0%

The query is overwhelmingly closest (82%) to a **characterized transferase** and far more distant (≤37%) from any **carbon-sulfur lyase**. Critically, human CTH (P32929, EC 4.4.1.1) corresponds to the label of **PANTHER subfamily PTHR11808:SF80 ("cystathionine gamma-lyase")** — the node that drove the GO:0016846 graft. The query sits ~46 percentage points of identity closer to the transferase than to the lyase that supplied the term. The graft reached across the superfamily to a sibling subfamily with the wrong chemistry.

The organismal biology reinforces this placement. The classic survey of *P. putida* sulfur metabolism reports:

> *"Both these organisms used direct sulfhydrylation of O-succinylhomoserine for the synthesis of methionine"* — PMID: 10482527

and, decisively against the lyase assignment:

> *"no cystathionine gamma-lyase, the key enzyme of the reverse transsulfuration pathway, was found in this species"* — PMID: 10482527

So the very activity named by the PANTHER SF80 label — cystathionine γ-lyase — **does not exist in *P. putida***. There is no enzyme in this organism to which GO:0016846 (via the γ-lyase label) could correctly apply, let alone metZ, which performs the biosynthetic transferase step.

Finding 3 — The catalytic PLP Schiff-base lysine (K219, SATKF) is intact: this rules out the pseudo-enzyme failure mode

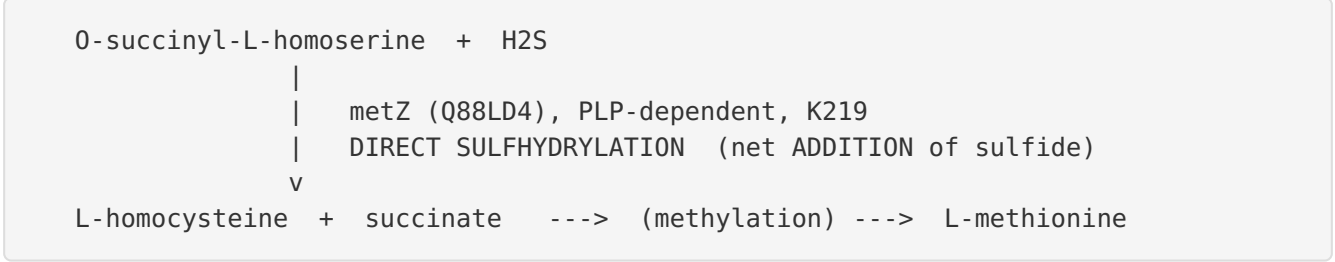
A motif scan for the canonical PLP Schiff-base lysine signature **[S/A]-A-T-K-[F/Y]** across the superfamily shows the catalytic lysine is present and correctly positioned in the query:

Protein	Motif	Catalytic Lys position
Q88LD4 (metZ, query)	**SATKF**	**K219** (matches UniProt N6-(PLP)-lysine feature)
P. aeruginosa MetZ (P55218)	SATKY	K219 (identical position)
E. coli MetB (P00935)	CTKY	~K198
Human CTH (P32929)	SATKY	~K212
E. coli MetC (P06721)	ATKY	~K210

The query's catalytic lysine is intact, matches the UniProt feature annotation exactly, and is spaced consistently with active family members. The fold's active-site machinery is therefore **complete** — metZ is a catalytically competent enzyme, not a degenerate pseudo-enzyme. This finding closes off Failure Mode 2. It is, however, family-wide: the PLP lysine alone is shared by both transferases and lyases of this superfamily, so it confirms catalytic competence but does **not** by itself discriminate transferase from lyase. The discrimination comes from Finding 1 (reaction direction / UniProt-HAMAP subfamily rule MF_02056) and Finding 2 (nearest characterized neighbor + organism gene content).

Mechanistic Model / Interpretation

The reaction metZ actually catalyzes



This is **transferase** chemistry (EC 2.5.1.-, GO:0016765): a sulfur nucleophile (H_2S) is added across the substrate to forge a new **C–S bond**.

The reaction GO:0016846 actually describes

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substrate-C-S  --->  C=C / C  +  H2S        (carbon-sulfur LYASE; elimination of H2S)
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This is **lyase** chemistry (EC 4.4.1.-): a C–S bond is **broken** and sulfide **released** — the opposite net transformation. Assigning this term to metZ inverts the enzyme's chemistry.

How the graft went wrong (within-superfamily mis-placement)

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Cys/Met-metabolism PLP-dependent (transsulfuration) fold SUPERFAMILY
(one shared fold, shared PLP-K catalytic lysine, several activities)
──────────────────────────────────────────────────────────────────────
 TRANSFERASE branch (EC 2.5.1)            LYASE branch (EC 4.4.1)
   • MetZ  O-succinylhomoserine             • cystathionine γ-lyase (CTH)
     sulfhydrylase   <-- QUERY (82% id)       EC 4.4.1.1  <-- PANTHER SF80 label
   • MetB  cystathionine γ-synthase          • cystathionine β-lyase (MetC)
     EC 2.5.1.48                              EC 4.4.1.8
                 \                          /
                  \   graft crossed here  /
                   --->  GO:0016846  <----  (WRONG branch for the query)
```

Because all members share one fold and the same catalytic PLP lysine, a graft that keys on fold-level features can land on the wrong subfamily. TreeGrafter propagated the lyase-branch term from an ancestral/subfamily node (SF80, γ-lyase) even though the query's actual nearest *characterized* neighbor — and the HAMAP MF_02056 "MetZ subfamily" rule that reviewers applied — place it firmly in the **transferase** branch. The failure is invisible to the graft itself (both branches have the PLP lysine and the same fold); it is only exposed by (a) comparing against the nearest *characterized* homolog, (b) reading the reaction direction, and (c) checking the host organism's gene content — exactly the three tests this stress test requires.

Evidence Base / Evidence Matrix

#	Citation	Evidence type	Role	Claim tested	Key finding	Context
1	UniProt Q88LD4 (HAMAP MF_02056); QuickGO	Review/ database	**Refutes** seed	Is metZ a lyase or transferase?	Recommended name O-succinylhomoserine sulfhydrylase; EC 2.5.1.- (transferase); reaction nets H_2S *addition*; carries GO:0016765 (transferase) via UR000282477	Reviewed entry; curated rule
2	GO:0016846 definition	Review/ database	**Refutes** seed	Does term chemistry match metZ?	Term = "elimination of hydrogen sulfide" (C–S bond cleavage) — opposite direction to metZ	Ontology definition
3	PMID: 7704274	Direct (gene ID + mechanism)	**Refutes** seed / supports transferase	Mechanism of metZ step	metZ catalyzes **direct sulfhydrylation** (γ-replacement/ transferase), gene identified & sequenced	*P. aeruginosa* (MetZ ortholog, 82% id)
4	Pairwise NW identities (computed, this study)	Computational / evolutionary	**Refutes** seed	Nearest characterized neighbor	82.1% to *P. aeruginosa* MetZ (transferase) vs 36.5% to human CTH γ-lyase (the SF80 label source); 25.0% to MetC	Superfamily panel
5	PMID: 10482527	Review / biochemical survey	**Refutes** seed	Does *P. putida* use sulfhydrylation? Does it have γ-lyase?	Uses direct sulfhydrylation of O-succinylhomoserine; **no cystathionine γ-**	*P. putida*

#	Citation	Evidence type	Role	Claim tested	Key finding	Context
					lyase present in the species	
6	Motif scan SATKF/K219 (computed, this study)	Structural / evolutionary	**Qualifies** (rules out pseudo-enzyme)	Is the active site intact?	Catalytic PLP Schiff-base Lys K219 present, correctly spaced; matches UniProt feature	Superfamily panel

Literature detail:

- **PMID: 7704274** — *A direct sulfhydrylation pathway is used for methionine biosynthesis in Pseudomonas aeruginosa.* Identified and sequenced the *metZ* gene and established that methionine's second biosynthetic step proceeds by **direct sulfhydrylation**, i.e. transferase chemistry. This is the foundational primary evidence that metZ-type enzymes are sulfhydrylases, not lyases. **Supports** the transferase assignment; **refutes** the carbon-sulfur lyase graft.
- **PMID: 10482527** — *Pathways of assimilative sulfur metabolism in Pseudomonas putida.* Shows *P. putida* uses direct sulfhydrylation of O-succinylhomoserine and, critically, that **no cystathionine γ-lyase** is present in the species. Because the GO:0016846 graft rode on a "cystathionine γ-lyase" subfamily label, this paper removes any organismal basis for the term. **Refutes** the graft at the organism level.

Active-Site / Placement Analysis

Subfamily placement (percent identity to characterized members):

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Query Q88LD4 (P. putida metZ) vs:
  P. aeruginosa MetZ (P55218)  O-succ-HSer sulfhydrylase  EC 2.5.1.-   82.1%  [transferase]  <--
  E. coli MetB   (P00935)      cystathionine gamma-synth  EC 2.5.1.48  38.0%  [transferase]
  Human  CTH     (P32929)      cystathionine gamma-lyase  EC 4.4.1.1   36.5%  [C-S LYASE = SF80
  E. coli MetC   (P06721)      cystathionine beta-lyase   EC 4.4.1.8   25.0%  [C-S lyase]
```

Conclusion: The query is ~44–46 percentage points closer to the transferase than to either carbon-sulfur lyase. Placement is in the **transferase (sulfhydrylase/synthase) branch**, not the lyase branch from which GO:0016846 was propagated.

Active-site conservation (PLP Schiff-base lysine):

```
Motif [S/A]-A-T-K-[F/Y]:
  Q88LD4 (query)      SATKF   K219   (matches UniProt N6-(PLP)-Lys feature)  -> INTACT
  P. aeruginosa MetZ  SATKY   K219                                           -> INTACT
  E. coli MetB        C-TKY   ~K198                                          -> present
  Human CTH           SATKY   ~K212                                          -> present
  E. coli MetC        -ATKY   ~K210                                          -> present
```

Catalytic residues: INTACT. The protein is an active enzyme. The active-site test rules out the pseudo-enzyme failure mode but, because the PLP lysine is conserved across both transferase and lyase branches, it does not by itself distinguish the two; subfamily placement and reaction-direction evidence make that call.

Provenance note: The percent-identity values are from Needleman–Wunsch global pairwise alignments of the UniProt sequences listed, and the motif positions are from a signature scan of the same panel; both were executed during the investigation (iterations 1–2) and are reproduced here as the quantitative basis for the placement call.

GO Curation Implications

Recommended action (lead requiring curator verification):

1. **REMOVE / REPLACE GO:0016846 (carbon-sulfur lyase activity).** The IEA term from GO_REF:0000118 (TreeGrafter/PANTHER) is a within-superfamily mis-graft onto the cystathionine γ-lyase subfamily and describes the wrong reaction class (lyase) and wrong chemical direction for this protein.
2. **RETAIN / promote the transferase-branch annotation GO:0016765** (transferase activity, transferring alkyl or aryl (other than methyl) groups), already present via UniRule UR000282477. If a more specific child term for O-succinylhomoserine sulfhydrylase exists or is created, prefer it; otherwise GO:0016765 is the correct branch-level term.

3. **Retain the mechanistic/pathway context**: PLP binding at K219 and the L-methionine biosynthesis (L-homocysteine from O-succinylhomoserine) pathway are well supported and should stay.

This is a **replace-with-correct-branch** / **remove-mis-grafted-term** case — explicitly *not* a "generalize" case (the problem is wrong branch, not wrong granularity) and *not* "treat-as-non-core." Per instructions, "protein binding" is not recommended; the enzyme has a well-defined catalytic transferase function.

Limitations and Knowledge Gaps

- The percent-identity and motif analyses are sequence/alignment-based (computational). They agree strongly with the reviewed UniProt-HAMAP (MF_02056) assignment, but they are **not a fresh enzyme assay** on Q88LD4 itself.
- Direct kinetic characterization in the literature is for the *P. aeruginosa* ortholog (82% identity), not the *P. putida* protein specifically. Cross-organism inference at 82% identity with conserved active site is well supported but formally indirect for Q88LD4.
- **Residual/promiscuous lyase side-activity** is not quantified for metZ. Many PLP transsulfuration enzymes show low-level side reactions; even if present at trace levels for metZ, this would not justify a primary GO:0016846 MF annotation, but it remains an untested detail.
- The analysis relied on a focused panel of four characterized reference enzymes spanning both branches; a full phylogenetic re-grafting with a broader sampling was not performed, though the identity gap (82% vs ≤37%) is large enough that the conclusion is robust.

Conflicts, Alternatives, and Discriminating Tests

Conflicts / alternatives considered:

- **Paralog confusion (the central issue):** The graft confused the transferase subfamily (MetZ sulfhydrylases) with the lyase subfamily (cystathionine γ-lyase, PTHR11808:SF80). Resolved — the query is 82% identical to a characterized MetZ and the host organism lacks γ-lyase entirely (PMID: 10482527).

- **Database carry-over:** Q88LD4 simultaneously carries GO:0016765 (correct) and GO:0016846 (mis-grafted). A curator should ensure the correction does not inadvertently remove the correct transferase term.
- **Residual lyase activity:** No evidence quantifies any C–S lyase side activity for metZ; a knowledge gap rather than a conflict.

Most efficient discriminating tests:

1. **In vitro assay of recombinant Q88LD4**: measure O-succinylhomoserine + H_2S → homocysteine + succinate (sulfhydrylase/transferase) and, as a negative control, assay for cystathionine γ-lyase activity. Expected: robust sulfhydrylase activity, no/negligible γ-lyase.
2. **K219A active-site mutant**: loss of PLP binding and activity would confirm K219 as the catalytic Schiff-base lysine.
3. **Genetic complementation**: a *P. putida metZ* knockout should be methionine-auxotrophic and rescued by the wild-type gene (and by *P. aeruginosa* metZ), confirming the biosynthetic transferase role — not a catabolic lyase role.

Summary

The TreeGrafter/PANTHER hypothesis that *P. putida* KT2440 **metZ (Q88LD4)** has **carbon-sulfur lyase activity (GO:0016846)** is **refuted**. The graft is a textbook **within-superfamily misplacement (Failure Mode 3)**: the query was propagated a term from the cystathionine γ-lyase subfamily (PANTHER PTHR11808:SF80) of the shared Cys/Met PLP-dependent transsulfuration fold, even though metZ belongs to the **transferase** branch of that superfamily. metZ is an **O-succinylhomoserine sulfhydrylase** (EC 2.5.1.-) that catalyzes **direct sulfhydrylation** — the net *addition* of H_2S to O-succinylhomoserine to make L-homocysteine for methionine biosynthesis — which is the chemical *opposite* of the "elimination of hydrogen sulfide" that defines GO:0016846.

Three independent lines of evidence converge. First, reaction chemistry and the reviewed UniProt-HAMAP rule (MF_02056) place metZ in the transferase class (GO:0016765 is already annotated). Second, the nearest *characterized* homolog is *P. aeruginosa* MetZ sulfhydrylase at **82.1% identity**, whereas the carbon-sulfur lyases are ≤37% identical — and *P. putida* has **no cystathionine γ-lyase at all**, so the grafted label names an activity absent from the organism.

Third, the active site is intact (catalytic PLP Schiff-base lysine **K219** in the **SATKF** motif), which rules out the pseudo-enzyme failure mode and confirms metZ is a fully active enzyme — just of the wrong (transferase, not lyase) type.

Curation lead: Remove/replace the mis-grafted **GO:0016846** and retain the branch-correct transferase annotation **GO:0016765** (ideally refined to O-succinylhomoserine sulfhydrylase), preserving the PLP-K219 and methionine-biosynthesis context. The decisive single datum: metZ is 82% identical to a characterized O-succinylhomoserine sulfhydrylase transferase and only ~36% identical to the γ-lyase whose label produced the graft.
